Question	Answer	Marks
1	**Defining the problem**	
	vary *C* and measure *e* or *C* is the independent variable and *e* is the dependent variable	**1**
	Keep *V* constant	**1**
	Methods of data collection	
	Labelled diagram of workable experiment including: • wooden block (and motor) fixed, e.g. G-clamp, • minimum two labels from method of fixing block (and motor) from G-clamp, block, spring, string, motor, spindle, bench.	**1**
	Voltmeter connected across capacitor or power supply in charging circuit	**1**
	Workable circuit diagram showing d.c. power supply and appropriate switches to allow for the charging of the capacitor **and** the discharging of the capacitor through the motor only **and** correct circuit symbol for capacitor	**1**
	Description of instrument to determine *e*, e.g. use rule(r) or calipers to measure length of spring	**1**
	Method of Analysis	
	Plots a graph of e^2 against *C* or *C* against e^2 or equivalent. (e.g. lg *e* against lg *C*)	**1**
	relationship valid if a straight line passing through the origin is drawn (For lg *e* against lg *C*: relationship valid if a straight line with gradient = 0.5 is drawn)	**1**
	e^2 against *C*: $\beta = \frac{k}{2V^2} \times \text{gradient}$ *C* against e^2: $\beta = \frac{k}{2V^2} \times \frac{1}{\text{gradient}}$ (e.g. lg *e* against lg *C*: $\beta = \frac{k}{2V^2} \times 10^{2 \times \text{y-intercept}}$)	**1**

Question	Answer	Marks
1	**Additional detail including safety considerations** Any **six** from:	**6**
	D1 precaution to prevent injury from moving springs, e.g. screens around apparatus in case spring snaps due to too much tension / goggles to prevent spring hitting face	
	D2 keep *k* constant	
	D3 description of method to determine *k*, e.g. add mass to spring and $k = mg$ / extension OR use newton-meter to measure force applied to spring and *k* = force / extension OR take several readings of force and extension, plot a force-extension graph and *k* = gradient	
	D4 method to determine *C*, e.g. time discharge of capacitor through a resistor to determine time constant or time for a change in current or potential difference	
	D5 description of method to determine *C* e.g. determine time constant from a graph **and** *C* = time constant / *R* **OR** determine values of *V* or I at different times **and** use of discharge of a capacitor equation	
	D6 Method to measure spring combination length, e.g. fix rule(r) to the bench or above springs OR add a pointer to the end of the spring combination to identify end-point of spring	
	D7 *e* = extended length of spring – original length of spring	
	D8 use a large value of *V* or small value of *k* to increase value of *e*	
	D9 repeat determination of *e* for the same value of *C* **and** determine the mean *e*	

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Question	Answer	Marks
1	D10 check the springs have not been stretched beyond their elastic limit or check that the springs return to their original lengths (when released / each time)	

Question	Answer	Marks
2(a)	Gradient = n y-intercept = $\lg\left(\sqrt{\frac{\gamma R}{M}}\right)$ **or** $0.5\lg\left(\frac{\gamma R}{M}\right)$	**1**
2(b)	**lg (v / m s^{-1})** 2.530 or 2.5302 ± 0.003 2.539 or 2.5391 ± 0.003 2.545 or 2.5453 ± 0.002 or ± 0.003 2.553 or 2.5527 ± 0.002 or ± 0.003 2.558 or 2.5575 ± 0.003 or ± 0.004 2.568 or 2.5682 ± 0.003 or ± 0.004	
	Values of lg (v / m s^{-1})	**1**
	Uncertainties in lg (v / m s^{-1})	**1**
2(c)(i)	Six points from (b) plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	**1**
	Error bars in lg (v / m s^{-1}) plotted correctly. All error bars to be plotted. Total length of bar must be accurate to less than half a small square and symmetrical.	**1**

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